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How to cite: *Angew. Chem. Int. Ed.* **2024**, *63*, e202410743
doi.org/10.1002/anie.202410743

Enantioconvergent Catalysis **Hot Paper**

Nickel-Catalyzed Enantioconvergent Allenylic Amination of Allenols Activated by Hydrogen-Bonding Interaction with Methanol

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Abstract: The ubiquitous nature of amines in drug compounds, bioactive molecules and natural products has fueled intense interest in their synthesis. Herein, we introduce a nickel-catalyzed enantioconvergent allenylic amination of methanol-activated allenols. This protocol affords a diverse array of functionalized allenylic amines in high yields and with excellent enantioselectivities. The synthetic potential of this method is demonstrated by employing bioactive amines as nucleophiles and conducting gram-scale reactions. Furthermore, mechanistic investigations and DFT calculations elucidate the role of methanol as an activator in the nickel-catalyzed reaction, facilitating the oxidative addition of the C–O bond of allenols through hydrogen-bonding interactions. The remarkable outcomes arise from a rapid racemization of allenols enabled by the nickel catalyst and from highly enantioselective dynamic kinetic asymmetric transformation of η^3-alkadienylnickel intermediates.

Introduction

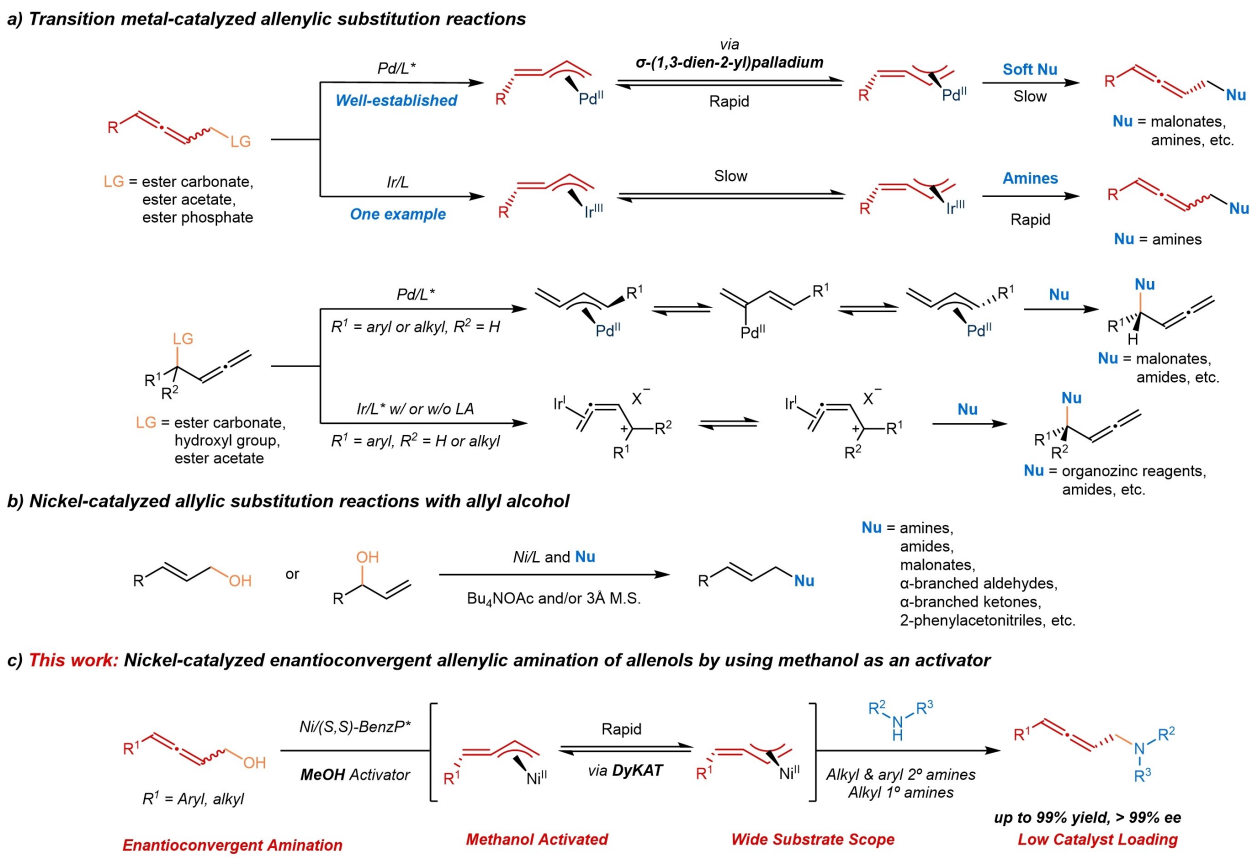
The amines are a critical component of a wide range of existing drug compounds, bioactive molecules, and anticancer agents, contributing significantly to their diverse biological and pharmacological activities. Remarkably, more than 80% of U.S. FDA-approved pharmaceuticals contain amine fragments.[1] Typical methods for incorporating an amine moiety into carbon chains include nucleophilic substitution reactions of amines with alkyl halides (or pseudohalides), reductive amination, and transition-metal-catalyzed C–N cross coupling reactions. Among these methods, transition-metal-catalyzed allylic amination has emerged as a widely utilized technique for the precise construction of C(sp^3)–N bonds with control over both regio- and stereochemistry. However, the relatively strong Lewis basicity of aliphatic amines leads to strong coordination with transition-metal catalysts in some cases, resulting in catalyst poisoning and the suppression of key elementary steps in the catalytic cycle. As such, enantioselective variants of allylic amination reactions employing aliphatic amines, enabled by transition-metal catalysis, have faced significant limitations in their development and success.[2] On a related note, while allenylic substitution reactions are conceptually similar to transition-metal-catalyzed allylic substitution, they are often more challenging to execute due to the formation of less reactive η^3-alkadienylmetal intermediates during the oxidative addition of metal catalysts to allenylic substrates. Palladium[3] and iridium[4] complexes are the most commonly used catalysts. Upon reacting with racemic allenylic substrates, a pair of diastereomeric η^3-alkadienyl-palladium complexes is formed. The two diastereomeric complexes can rapidly interconvert to each other through a η^1-alkadienylmetal complex intermediate, allowing a subsequent attack by a nucleophile to give an enantioconvergent allenylic product.[3a–c,e,g–k,n] In contrast, in a similar catalytic process mediated by an iridium catalyst, the diastereomeric η^3-alkadienyliridium intermediates are captured by the nucleophiles prior to interconversion, thus leading to allenylation products with retention of configuration and a kinetic resolution of racemic allenylic reactants.[4] In addition, either palladium[5] or iridium[6] complexes could also enable α-branched racemic allenylic substrates to undergo stereoconvergent allenylic substitution reactions (Scheme 1a). Based on these general principles, a variety of enantioselective allenylic substitution reactions with a broad range of soft nucleophiles have been developed.[3b–e,g–k,m–p,5–6] However, there is only a limited number of reports on asymmetric allenylic amination between allenylic phosphates or carbonates and amines or amides enabled by chiral palladium complex catalysis.[3c,d,g,h,5b,6c] On another note, nickel, which features higher natural abundance, chemical sustainability, and affordability compared to noble metals, has emerged as an important alternative in catalysis over last several decades.[7] Notably, nickel complexes allow allylic alcohols to directly participate in asymmetric allylic alkylation reactions without the need for external activators of hydroxy groups (Scheme 1b).[8] This feature confers both step- and atom-economy benefits. Surprisingly, nickel-catalyzed allenylic substitution reactions to access axial products have not yet been reported,[9] despite the well-established

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Scheme 1. Transition metal-catalyzed allenylic substitution reactions.

success of nickel in allylic substitution reactions over many years.[8,10]

In this study, we present the first nickel-catalyzed enantioconvergent allenylic amination of allenols with a broad substrate scope, accommodating a diverse range of allenols and amines. The reaction is compatible with a multitude of functional groups, drug fragments, and natural products, yielding an extensive collection of allenylic amines characterized by near-perfect yields and enantioselectivities (Scheme 1c). Furthermore, mechanistic studies and density functional theory (DFT) calculations indicate that a rapid interchange between diastereomeric η^3-alkadienylnickel intermediates allows an efficient dynamic kinetic asymmetric transformation (DyKAT) that turns out to be the origin of high enantioselectivity. In addition, the calculations suggest that the activation of allenols is promoted by a hydrogen-bonding interaction with methanol molecule, aiding the oxidative addition of the nickel catalyst. Subsequently, the resulting η^3-alkadienylnickel intermediate undergoes amination through an outer-sphere amine-attack pathway.

Results and Discussion

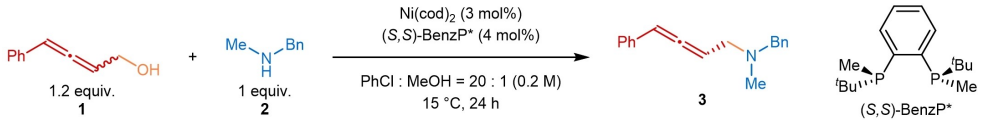
Our initial research efforts were concentrated on optimizing the nickel-catalyzed asymmetric allenylic amination between a racemic allenol **1** and *N*-methylbenzyl amine **2**. A systematic examination of the reaction parameters and ligands identified a chiral nickel complex in situ formed from $Ni(cod)_2$ and (*S,S*)-BenzP* to be the optimal catalyst (see Tables S1–S4 in the Supporting Information for more details). In the presence of 3 mol % $Ni(cod)_2$ and 4 mol % (*S,S*)-BenzP* and in a solvent mixture of chlorobenzene and methanol, the reaction proceeded smoothly to afford the desired allenylation product **3** in >95 % NMR yield and with 99 % ee (entry 1, Table 1). As anticipated, no reaction was detected in the absence of the nickel catalyst (entry 2). Interestingly, the reaction got much slower in the absence of methanol, indicating that methanol is probably an external activator for the hydroxyl group through a hydrogen-bonding interaction (entry 3).[11] However, the presence of chiral phosphoric acid was unable to speed up the reaction while obvious difference in the stereochemical control was found between (*S*) and (*R*)-TRIP[12] in chlorobenzene at 15 °C (entries 4 and 5). The choice of nickel precursor also had a profound impact on the reaction. Neither air-stable nickel(0) nor nickel(II) pre-catalysts catalyzed this reaction (entry 6). A minor modification in the ratio of **1** to **2** adversely affected the yield (entry 7). Adjusting either the solvent or the nickel-to-ligand ratio failed to enhance the reaction performance (entries 8 and 9).

With the optimized reaction conditions in hand, our initial assessment aimed to explore the versatility of various amines in conjunction with racemic 4-phenylbuta-2,3-dien-1-

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Table 1: Optimization Studies.[a]



entry	variations	yield/%	ee/%
1	None	>95	99
2	Without $Ni(cod)_2$ and ligand	N.R.	–
3	Without MeOH	29	98
4	10 mol% (*S*)-TRIP instead of MeOH	28	98
5	10 mol% (*R*)-TRIP instead of MeOH	31	88
6	Ni(cod)DQ or $NiBr_2 \cdot$DME instead of $Ni(cod)_2$	N.R.	–
7	**1**:**2** = 1:1.2	66	98
8	PhMe instead of PhCl	74	98
9	[Ni]:Ligand = 1:1 or [Ni]:Ligand = 1:2	88/70	98/99

[a] Reactions were run on a 0.1 mmol scale of **2**. The yield of **3** was determined by ^{1}H NMR analysis of the crude reaction mixture using the CH_2Br_2 as an internal standard. The ee was determined by HPLC.

ol **1**. As demonstrated in Scheme 2, a wide range of amines showcased their capacity to participate in this protocol, furnishing the corresponding (*S*)-allenylic amines in moderate to good yields, with the ee values mostly exceeding 97%. Linear secondary amines (**3**–**7**) were well-tolerated, while increased steric hindrance of amines resulting in significantly reduced yields. Cyclic aliphatic amines of varying ring sizes, encompassing azetidine, pyrrolidine, piperidine, 4-phenylpiperidine, azepane, and azocane (**8**–**13**), proved to be efficient nucleophiles, with particularly noteworthy ee values for piperidine or azocane (98%). Furthermore, a spectrum of six-membered heterocyclic amines (**14**–**22**) smoothly underwent transformation, consistently yielding products with >97% ee. Among these heterocyclic amines, both morpholine and *N*-substituted piperazine are frequently encountered in drug molecules.[1b] Cyclic aromatic amines, such as indoline, also exhibited sufficient nucleophilicity to engage in the allenylic amination reaction (**23**). However, broadening the scope of the reaction to primary amines, such as benzylamine, resulted in a significantly poorer outcome, giving **24** with mere 15% yield and 66% ee. Nevertheless, after another systematic optimization of reaction conditions (see Tables S5 and S6 in the Supporting Information for more details), **24** could be furnished with synthetically valuable 54% yield and 94% ee. However, anilines showed very poor reactivity, leading to trace amounts of product presumably due to the low nucleophilicity (see Supporting Information).

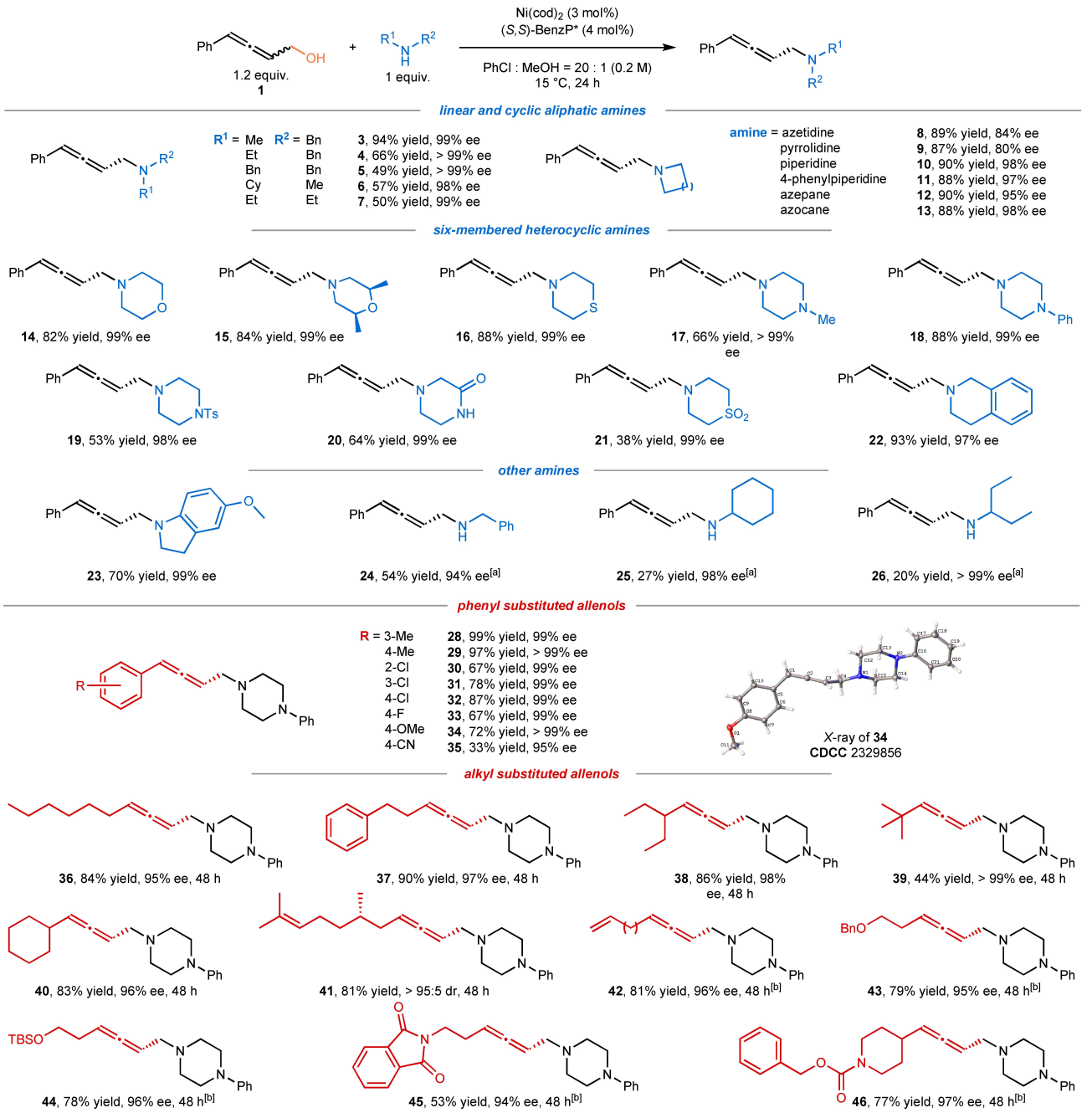
We then shifted our focus toward exploring the range of racemic allenols through the reaction with 1-phenylpiperazine **27**. Various substituted phenyl groups bearing either electron-donating or electron-withdrawing substituents (**28**–**35**) were found to be compatible. In general, very high enantioselectivity was achieved in all cases (95 to >99% ee), although the yield appeared to correlate with the electron density and steric hindrance of the phenyl group. Consequently, the reaction with **35**, bearing a *para*-cyanophenyl, provided a considerably lower yield. Importantly, the reactions involving allenols substituted with alkyl groups bearing primary, secondary, tertiary carbon (**36**–**39**) and cyclic secondary carbon **40** all underwent the reaction smoothly. Although only a moderate yield was obtained for *t*Bu-substituted allenol, it features the highest enantioselectivity (>99% ee). More importantly, allenols carrying alkyl groups with synthetically valuable functional groups, such as internal alkene or terminal alkene (**41** and **42**), benzyl or silyl ethers (**43** and **44**) and amides (**45** and **46**), proved to be suitable reactants, furnishing multiply functionalized allenylic amines with excellent enantioselectivities (94–97% ee). The absolute configuration of **34** was unambiguously determined through X-ray crystallographic analysis.[13]

We next assessed the applicability of this allenylic amination process to the late-stage functionalization of structurally complex molecules (Scheme 3). A dozen of drugs, drug fragments, and natural products (**48**–**59**), featuring diverse motifs, were successfully modified into their corresponding allenylic amines in good yields and with excellent stereoselectivities.[14] Notably, this protocol provides an efficient route to the optically pure allenylic analog of naftifine **48**. Particularly remarkable was the nearly quantitative transformation of much more stable amine hydrochlorides (**53**–**56**) into allenylic amines with excellent enantioselectivity in the presence of Cs_2CO_3. To further demonstrate the synthetic applicability, we tested gram-scale reactions employing only 2 mol% of the chiral nickel catalyst, which afforded the desired allenylic amines, maintaining the yield and enantioselectivity almost at par with the small-scale reaction.

To gain insight into the role of methanol in the activation of allenol, a series of control experiments were conducted. As shown in Scheme 4a, the addition of methanol obviously accelerated the reaction (entries 1–3).[15] Given that the high dielectric constant of polar solvent can facilitate reactions involving ionic intermediates,[16] it became necessary to ascertain whether the observed acceleration stems from the high polarity of methanol or from more specific interactions. Thus, a model reaction between **1** and **2** was conducted in a solvent mixture of chlorobenzene and dichloromethane (1.52:1 volume ratio), which should have the same dielectric constant as the solvent mixture

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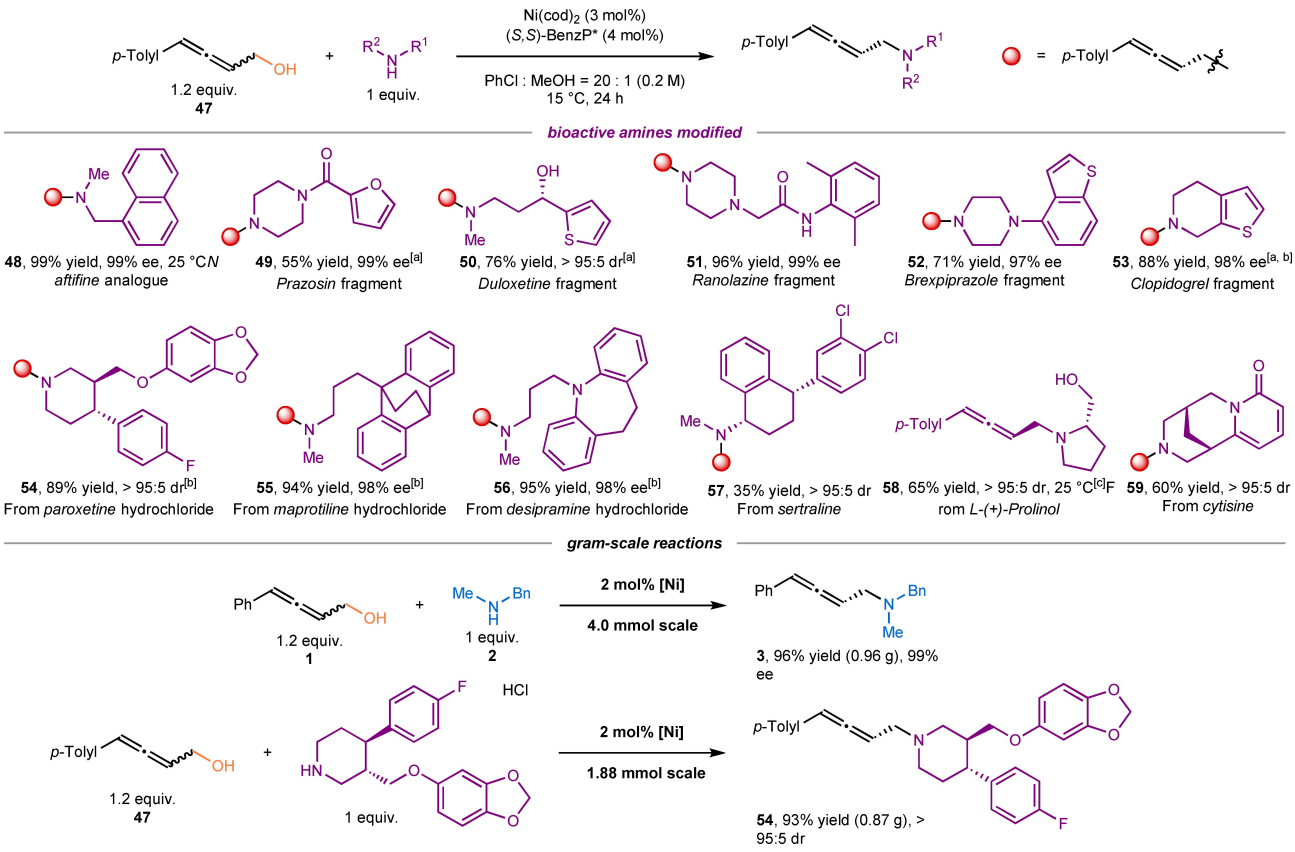


Scheme 2. Substrate scope of allenols and amines. Unless otherwise noted, the reactions were conducted on 0.4 mmol. The reported yield represents the isolated yield, enantiomeric excess (ee) was determined by HPLC, and the diastereomeric ratio (dr) was determined by ^{1}H NMR. [a] The reaction was run under the following condition for 24 hours: 1.0 equiv. **1**, 5.0 equiv. primary amine, 3 mol% $Ni(cod)_2$, 6 mol% (*S,S*)-BenzP* and solvent PhCl : MeOH = 10 : 1 (v : v). [b] 0.3 mmol scale.

employed under optimal conditions (chlorobenzene to methanol, 20 : 1 volume ratio), as per theoretical calculations ($\varepsilon = 6.980$ vs. $\varepsilon = 6.979$, detailed in Table S7 in the Supporting Information). However, in this solvent mixture, a low conversion was observed (entry 4), indicating that the solvent polarity has minimal impact on the reaction. This observation hints at more specific solvent interactions, with hydrogen bonding between methanol and allenols potentially facilitating the C–O bond-breaking event. This hypothesis was supported by DFT calculations, as discussed later. In addition, to identify if a kinetic resolution process happens, optically pure (*R*)- and (*S*)-**1** were examined as the substrates. However, both underwent the asymmetric allenylic amination reaction smoothly, giving rise to products with the same configuration, in similar yields and with identical enantioselectivity, showing the enantioconvergent nature of this reaction. Finally, the enantiomeric excess of 4-phenylbuta-2,3-dien-1-ol **1** was monitored during the reac-

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Scheme 3. Bioactive amines modified and gram-scale reactions. Unless otherwise noted, the reactions were conducted on the 0.3 mmol scale. The reported yield represents the isolated yield, the enantiomeric excess (ee) was determined by HPLC, and the diastereomeric ratio (dr) was determined by ^{1}H NMR. [a] 0.4 mmol scale. [b] Using amine hydrochlorides as nucleophiles and 1.1 equiv. Cs_2CO_3 as base. [c] (*R,R*)-BenzP* instead of (*S,S*)-BenzP*.

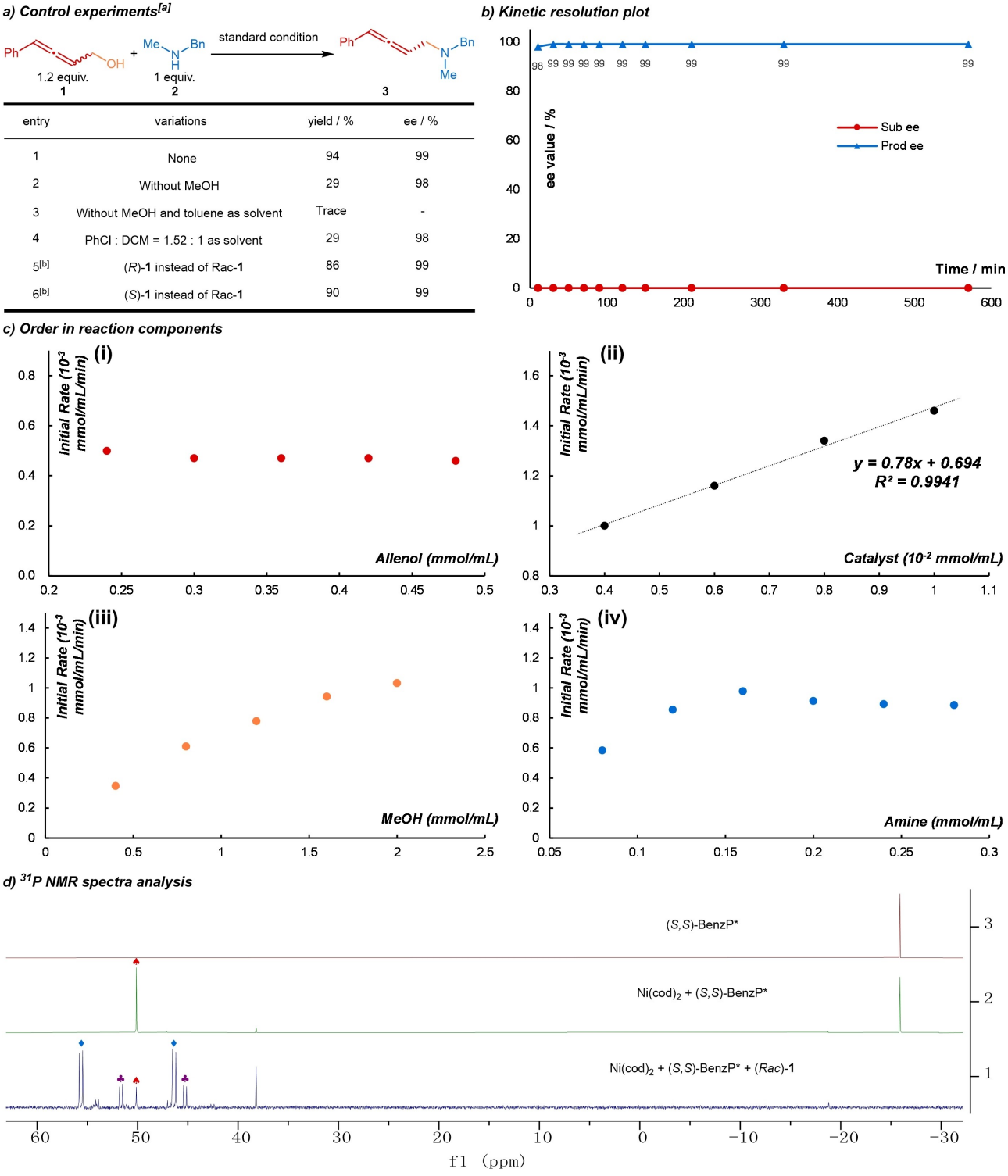
tion process. The remaining unreacted **1** was always racemic (<3% ee) throughout the reaction (Scheme 4b), supporting the notion that a kinetic resolution process does not take place, but instead, a rapid DyKAT occurs.

Moreover, kinetic order of each of reaction components, including catalyst, was measured by means of differential rate laws (Scheme 4c). A zero-order reaction was identified for the allenol **1** while a first-order reaction was observed for nickel catalyst, indicating that the oxidative addition of nickel complex to allenol **1** may not be the rate-limiting step. Interestingly, the reaction order for methanol is 0.68, implying that the methanol indeed participates in the important reaction steps, but plays a quite complicated role, for instance, it might form hydrogen-bonding network to exert impact on the reaction rate and stereochemistry.[10t,17] Even more interesting is that the reaction order for the amine nucleophile seemingly depends on the concentration.[18] The reaction gets gradually faster with increasing the concentration of the amine **3** from 0.08 M to 0.16 M, but in contrast, the rate becomes slower at higher concentration as shown in the reaction profile with the concentration of **3** from 0.16 M to 0.28 M. The high concentrated amine slowing down the reaction might be arisen from the deactivation of allenol by forming a hydrogen-bond with methanol to somewhat inhibit the hydrogen-bonding interaction with allenol (titration experiments of **1** with MeOH and **2** with MeOH were shown in the Supporting Information, Figures S6–7 and S8–9). ^{31}P NMR spectroscopic analysis of the reaction mixture was performed to identify the key intermediates (Scheme 4d). In comparison with the ^{31}P NMR data of (*S,S*)-BenzP*, the 1:1 mixture of $Ni(cod)_2$ and (*S,S*)-BenzP* in d_8-toluene generates a new peak at 50.1 ppm, which can be assigned to the complex of nickel and chiral ligand. After addition of one equivalent of the allenol **1** to above-mentioned solution, two sets of doublets with a ratio of 2.5:1 in intensity at 55.8, 51.8 ppm and 46.5, 45.4 ppm, respectively, which could be tentatively assigned to diastereomeric mixture of Ni[(*S*)-**1**][(*S,S*)-BenzP*] and Ni[(*R*)-**1**][(*S,S*)-BenzP*], suggesting that the kinetic resolution in the coordination of chiral nickel catalyst and racemic allenol **3** occurs. However, since the oxidative addition is not rate-limiting step (Scheme 4c), the whole catalytic cycle does not result in kinetic resolution, that is, the recovered allenol **3** remains racemic.

To identify the origin of the high ee values observed in the reactions and unravel the underlying reaction mechanism, DFT calculations were performed on potential intermediates and transition states within the catalytic cycle (Scheme 5a). The hydrogen bond titration experiment demonstrates that both methanol and the allenol can form

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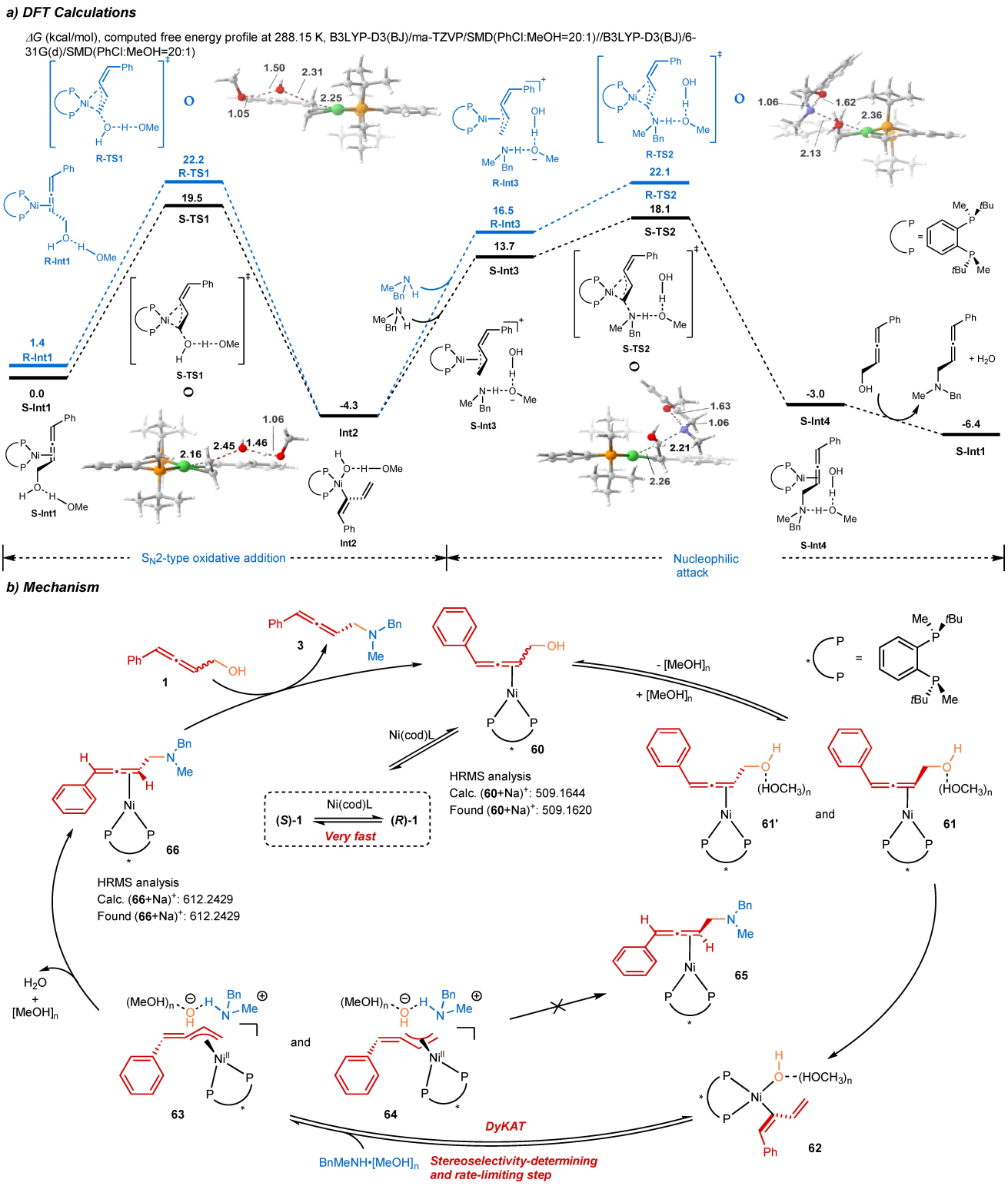


Scheme 4. Mechanistic studies. [a] Reactions were run using a 0.1 mmol scale of **2** at standard condition. The yield of **3** was determined by ^{1}H NMR analysis of the crude reaction mixture using the CH_2Br_2 as an internal standard. The ee was determined by HPLC. [b] The ee for both (*R*)-**1** and (*S*)-**1** was 97%.

complex hydrogen bond networks, however, for computational purposes, we simplify it to a single methanol model here, considering that the kinetic order of methanol is not greater than 1 (Scheme 4c-iii). The catalytic cycle initiates with Ni complexes, **S-Int1** and **R-Int1**, generated from a complexation of Ni catalyst and racemic allenol substrate **1**, respectively, while forming hydrogen bond with one molecule of methanol. The DFT results show that **S-Int1** is more

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Scheme 5. Computational studies and reaction mechanism.

stable than **R-Int1**, which is consistent with the NMR results (Scheme 4d). Both **S-Int1** and **R-Int1** can undergo S_N2-type oxidative addition via **S-TS1** and **R-TS1**. Subsequently, the oxidative addition product quickly transforms into a more stable η^1-(1,3-dien-2-yl) nickel species **Int2**. Afterwards, the η^1-1,3-diene-2-yl nickel species **Int2** isomerizes again into the η^3 species and interacts with amines to form **S-Int3** and **R-Int3**, respectively. Due to the difference in stability between **S-Int3** and **R-Int3**, **S-Int3** is more likely to be formed. Subsequently, the amine undergoes nucleophilic attack

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through **S-TS2** and **R-TS2** (**S-TS2** 22.4 kcal/mol vs **R-TS2** 26.4 kcal/mol). It is worth mentioning that the formation of hydrogen bonds between the methoxy anion and the amine enhances the nucleophilic attack ability of the amine if methanol is present in the system. This process is also an enantio-determining step, as the **R-TS2** barrier is much higher than **S-TS2**, leads to the high ee values. Additionally, the reversibility of the isomerization process in the preceding step enables the conversion of **R-Int3** to **S-Int3** by reverting to the **Int2**, thus sustaining the reaction and facilitating an overall reaction yield exceeding 50 %. Finally, the catalyst is regenerated and the product is released.

Based on the experimental findings, prior reports on related transitional-metal-catalyzed reactions,[3a–c,e,g–k,n,8–9] and DFT computational results, we propose a reaction mechanism (Scheme 5b). The coordination of allenol **1** with the chiral nickel complex leads to the formation of [(η^2-allenol)Ni((*S,S*)-BenzP*)] **60**. Subsequently, methanol molecules form hydrogen bonds with the hydroxyl group of allenol **1**.[11a,19] Such hydrogen-bonding interactions activate the allenylic C–O bond and facilitate the oxidative addition of the nickel complex **61**, giving rise to η^1-(1,3-dien-2-yl)nickel intermediate **62**. Simultaneously, a nickel complex **61′**, the diastereomer of **61**, might undergo a rapid isomerization to go back to **1** as indicated by control experiments (Tables S14 and S15 in Supporting Information). Subsequently, amine **2**, activated by the hydroxide and stabilized through a hydrogen-bonding interaction with methanol, approaches the η^1-(1,3-dien-2-yl)nickel intermediate (**62**) to form complexes **63** and **64**. The rapid interconversion between these intermediates enables an efficient DyKAT via the η^1-(1,3-dien-2-yl)nickel intermediate **62**. While outer-sphere nucleophilic attack of the amine **2** on the η^3-phenylmethylene-π-allylnickel intermediate **63** results in the irreversible formation of a relatively more stable intermediate **66**, **64** encounters significant steric repulsion between amine **2** and the *t*Bu group on the ligand, making it difficult to undergo C–N coupling and yield **65**. Finally, the product **3** is released through a substrate substitution reaction of the intermediate **66** with the allenol **1**, concomitant with the regeneration of the catalyst [(η^2-allenol)Ni((*S,S*)-BenzP*)] (**60**). In this catalytic cycle, putative intermediates **60** and **66** were identified through HRMS analysis.

Conclusion

In summary, we have developed a nickel-catalyzed enantioconvergent allenylic amination of allenols, offering an efficient route to a wide array of densely functionalized allenylic amines in high yields and with excellent enantioselectivities. Notably, this method allows for the incorporation of bioactive amine fragment derivatives. Through mechanistic studies and DFT calculations, we have unveiled the pivotal role of methanol in activating the hydroxyl group, leading to a rapid DyKAT process during the formation of η^3-alkadienylnickel intermediates. This enables the stereoconvergent conversion of racemic allenols. The cooperative activation of Ni/hydrogen-bonding interaction represents a promising strategy for allenylic substitution reactions involving other nucleophiles beyond amines. The future exploration of this field holds great promise.

Acknowledgements

The financial support from the National Key R&D Program of China (2023YFA1506700) is gratefully acknowledged. We also thank Prof. Qing-Wei Zhang (USTC) and Dr. Elvis Wang Hei Ng (HKU) for insightful suggestions.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: enantioconvergent amination · allenylic substitution · nickel catalysis · hydrogen-bonding activation

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Manuscript received: June 7, 2024
Accepted manuscript online: July 4, 2024
Version of record online: August 21, 2024

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